

SEONVIN CHO

Integrated M.S./Ph.D. Student in Electrical Engineering, Hanyang University
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Research Interests

Reinforcement Learning · Sequential Decision Making · Generative Models

Education

Hanyang University, Seoul, Republic of Korea Sep. 2024 – Present

Integrated M.S./Ph.D. Student in Electrical Engineering

Advisor: Prof. Songnam Hong

GPA: 4.45 / 4.50

Hanyang University, Seoul, Republic of Korea Mar. 2020 – Aug. 2024

B.S. in Electrical Engineering

Summa Cum Laude, Early Graduation

GPA: 4.26 / 4.50

Research

PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning 2026

Soohyun Choi*, Seonvin Cho*, Songnam Hong[†] · Under Review · *Equal contribution · [†]Corresponding author

[Code] [arXiv]

- Developed a hierarchical offline goal-conditioned RL framework that connects subgoal selection to short-horizon execution through explicit state-space bridges.
- Combined generative endpoint proposals, transitive value-based selection, and inverse-dynamics decoding to produce executable short-horizon action chunks.

Federated Learning under Noisy Labels 2023

Undergraduate Capstone Project

[Code] [Technical Report]

- Analyzed loss-value distributions in federated learning to identify unreliable samples under label noise.
- Received the Best Award (Department) and Excellence Award (College of Engineering).

Multi-Step Proximal Policy Improvement Present

Ongoing Research

- Investigating re-centered multi-step proximal refinement that decouples critic bootstrap exposure from final-policy reach and improves stability across policy-improvement budgets.

Adaptive Policy Optimization Present

Ongoing Research

- Developing adaptive refinement of step size and depth, together with value-guided interpolation between behavior-supported student policies and multi-step teacher policies.

Extensions of PathBridger Present

Ongoing Research

- Exploring offline-to-online learning and more flexible long-horizon control.
- Extending state-space planning and generative approaches to action-free settings, where the data do not provide actions.

Honors & Awards

BK Research Assistantship (BK-RA), Hanyang University Spring & Fall 2026

IEEE Antennas and Propagation Society Undergraduate Summer Research Scholarship 2024

Excellence Paper Award, KIEES [Code] Winter & Fall 2024

Best & Excellence Awards, Federated Learning Capstone Project 2023

Excellence Award, Embedded Creative Robot Challenge 2022

Merit-Based & Academic Scholarships, Hanyang University 2021 – 2024